Power Electronics Onboarding

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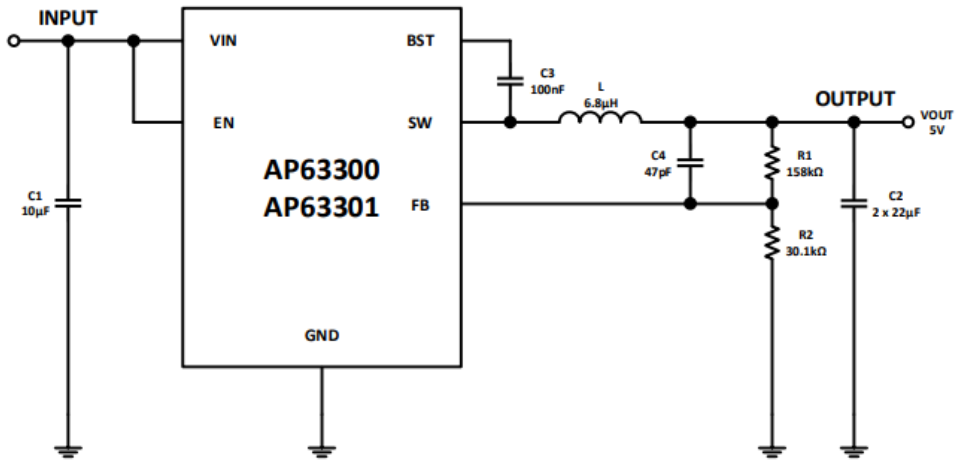


Figure 1. Typical Application Circuit

IC and Circuit details

I chose to build my circuit based off of this schematic. For the buck converter IC I chose to go with the AP63301WU-7. This IC has an output current of 3A, has a voltage rating from 3.8V to 32V and is able to output a voltage from 0.8V to 31V. It also has a frequency switching of 500kHz. The AP63301WU-7 has 6 pins FB, EN, VIN , GND , SW, and BST. The first pin FB is used to set the output voltage of the IC. Since the AP63301WU-7 has adjustable voltage, a voltage divider is connected between the output, FB pin, and ground, in order to get the desired output voltage. The voltage divider converts the output voltage into 0.8V, the target voltage value of the FB pin. Whenever the FB pin drops from 0.8V the IC increases its switching activity to raise output voltage. To decrease output voltage it decreases switching activity. In addition, a capacitor can be placed in parallel with resistor one in order to improve how the circuit recovers from sudden changes in current. The EN pin is used to switch on the AP63301WU-7 as the VIN increases. It can be connected to the VIN pin or left alone to turn on the IC. The VIN pin is used to connect the power source to the IC. A capacitor connected in between VIN and ground is used in order to eliminate noise from the switching of the IC. SW is used to supply power to the output. The inductor and capacitor make an LC filter in which the inductor minimizes current ripple and the capacitor reduces voltage ripple so the buck converter can produce a stable 5V DC output. The GND is used to power ground. The final pin is BST; it is used to supply voltage to Q1 (high-side mosfet) so it can turn on. If the capacitor is below 2.3V the IC switches to Q2 (low-side mosfet) for 220 nanoseconds to recharge it. This allows Q1 to have gate-driving voltage during light-load or low-dropout conditions.

Table 1. Recommended Component Selections

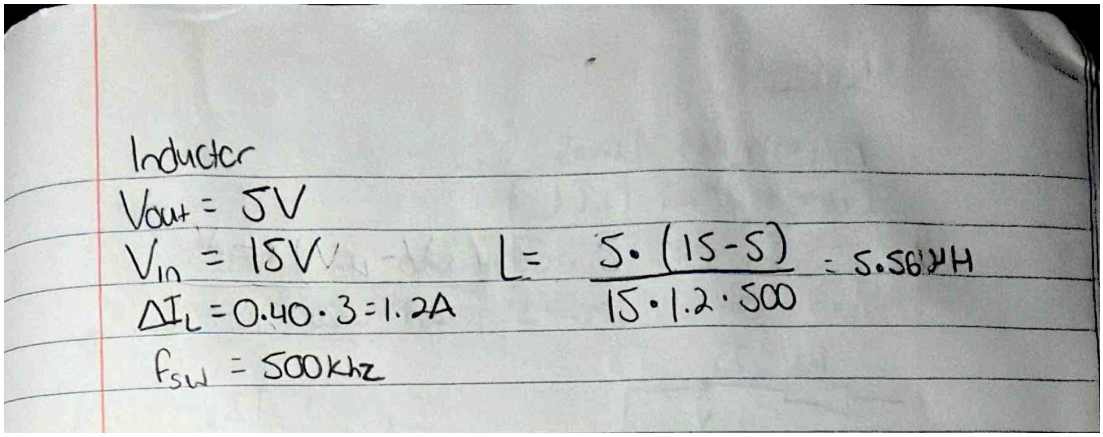
AP63300/AP63301							
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)
1.2	15.0	30.1	2.2	10	3 x 22	100	OPEN
1.5	26.1	30.1	3.3	10	3 x 22	100	OPEN
1.8	37.4	30.1	3.3	10	2 x 22	100	OPEN
2.5	63.4	30.1	4.7	10	2 x 22	100	56
3.3	93.1	30.1	4.7	10	2 x 22	100	56
5.0	158.0	30.1	6.8	10	2 x 22	100	47
12.0	422.0	30.1	10.0	10	2 x 22	100	12

The manufacturer of the AP63301WU-7 provided a list of recommended values to use for components. I followed all of these values, but had to add extra capacitors due to DC Bias.

Inductor

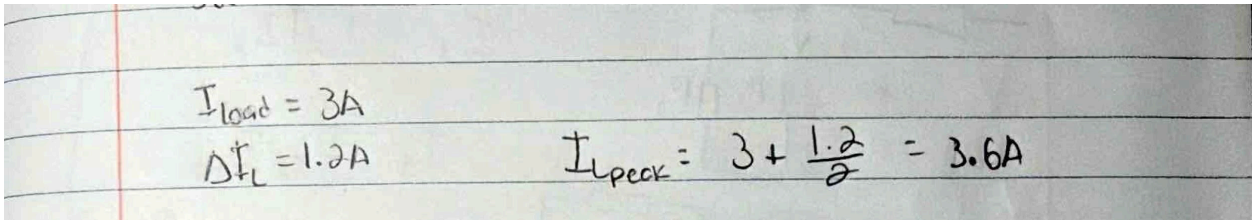
The inductor's main job in this circuit is to limit current ripple by resisting sudden changes in current. This allows for the output to receive the same current consistently. The Inductor I went with was the SRN8040TA-6R8M, a 6.8 µH inductor. I chose it based on the following criterias: current rating, current saturation, DC resistance, and inductance.

$$\mathbf{L = \frac{VOUT \cdot (VIN - VOUT)}{VIN \cdot \Delta I_L \cdot f_{sw}}}$$



Using the formula above I was able to calculate the required inductance I needed for my circuit. I got 5.56 µH but chose to use the recommended value of 6.8 µH on the data sheet.

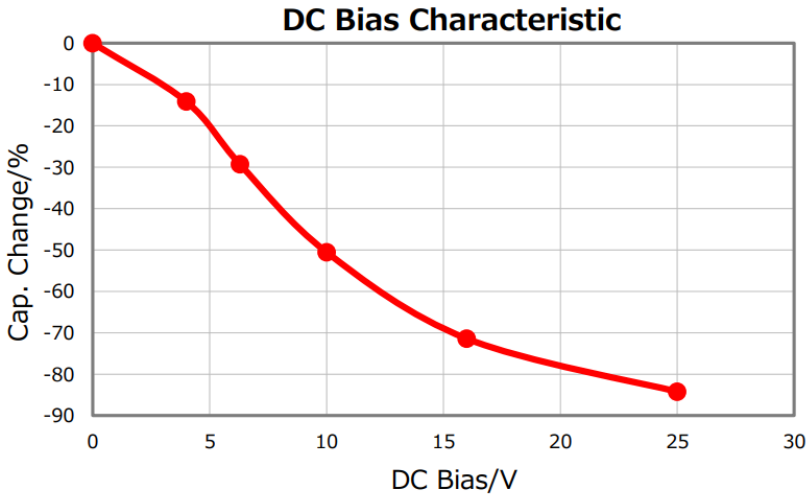
$$\mathbf{I_{L_{PEAK}} = I_{LOAD} + \frac{\Delta I_L}{2}}$$



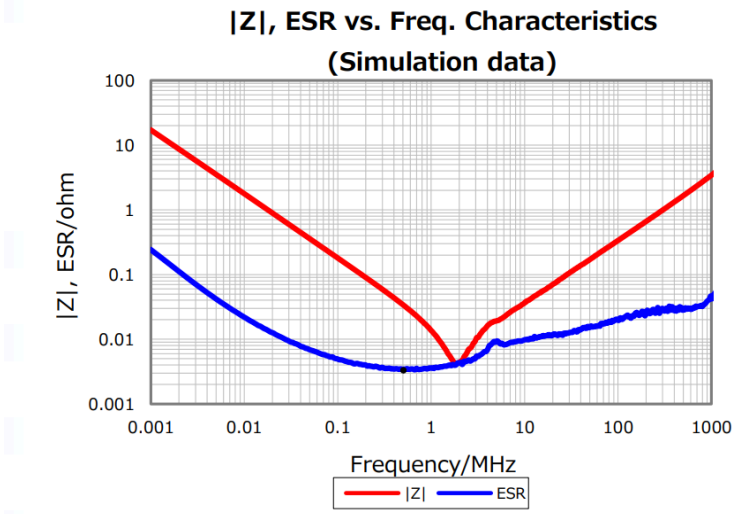
I calculated the peak current in order to see what current saturation value my inductor had to withstand. The SRN8040TA-6R8M has a saturation of 5.6A much higher than the minimum 3.6A. In addition, I also had to choose an inductor that could withstand a DC current rating of at least 35% or higher of the maximum load current which is 4.05A. The SRN8040TA-6R8M maximum load current is 5.1A. The inductor's DC resistance had to be less than 30mΩ. The SRN8040TA-6R8M has a maximum DC resistance of 25mΩ.

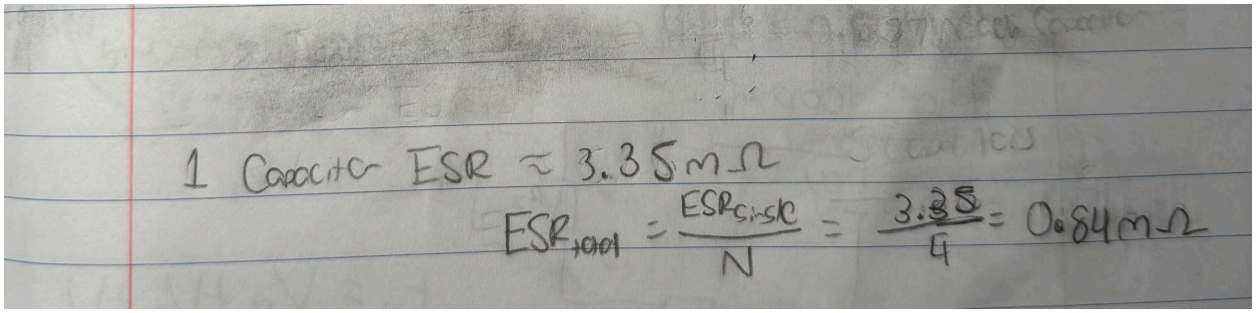
Input Capacitor

The input capacitor's main job is to reduce input voltage ripple and electrical noise from the ICs constant high frequency switching. It provides current to the IC which stabilizes input voltage. For the input capacitor I chose to go with the C2012X7S1E106K125AE. This is a ceramic capacitor with a rating of 10 µF and is able to withstand 25V. I chose this capacitor based on the DC Bias, rated voltage, RMS input current, and the ESR.

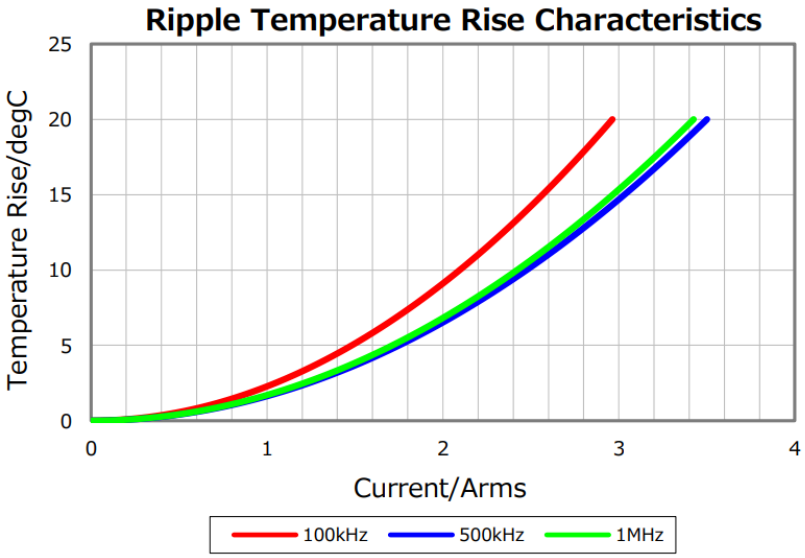


The DC Bias at 15V for the C2012X7S1E106K125AE is about 70%. Which makes one capacitor have a capacitance of about 3 µF at 15V. The circuit requires at least 10 µF of capacitance so putting four C2012X7S1E106K125AE in parallel allows for the minimum 10 µF goal to be achieved with 10% margin of error.





The manufacturer also requires a capacitor to have a low ESR due to large change of current. Using four capacitors in parallel allows the ESR to be divided in four reducing noise. For one capacitor the ESR was about 3.35mΩ. With four capacitors in parallel this becomes 0.84mΩ.

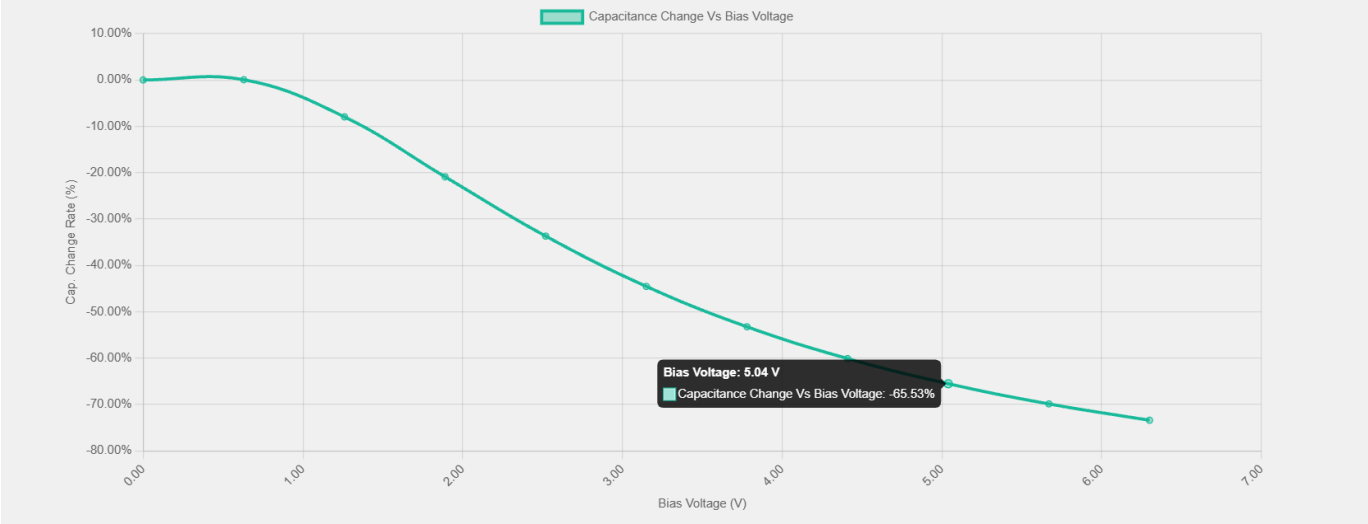


X

The total Arms of four capacitors in parallel at 500kHz is 0.346A. For each individual capacitor it is 0.087A. Based on the graph the current produces a small temperature increase making it safe to use these capacitors.

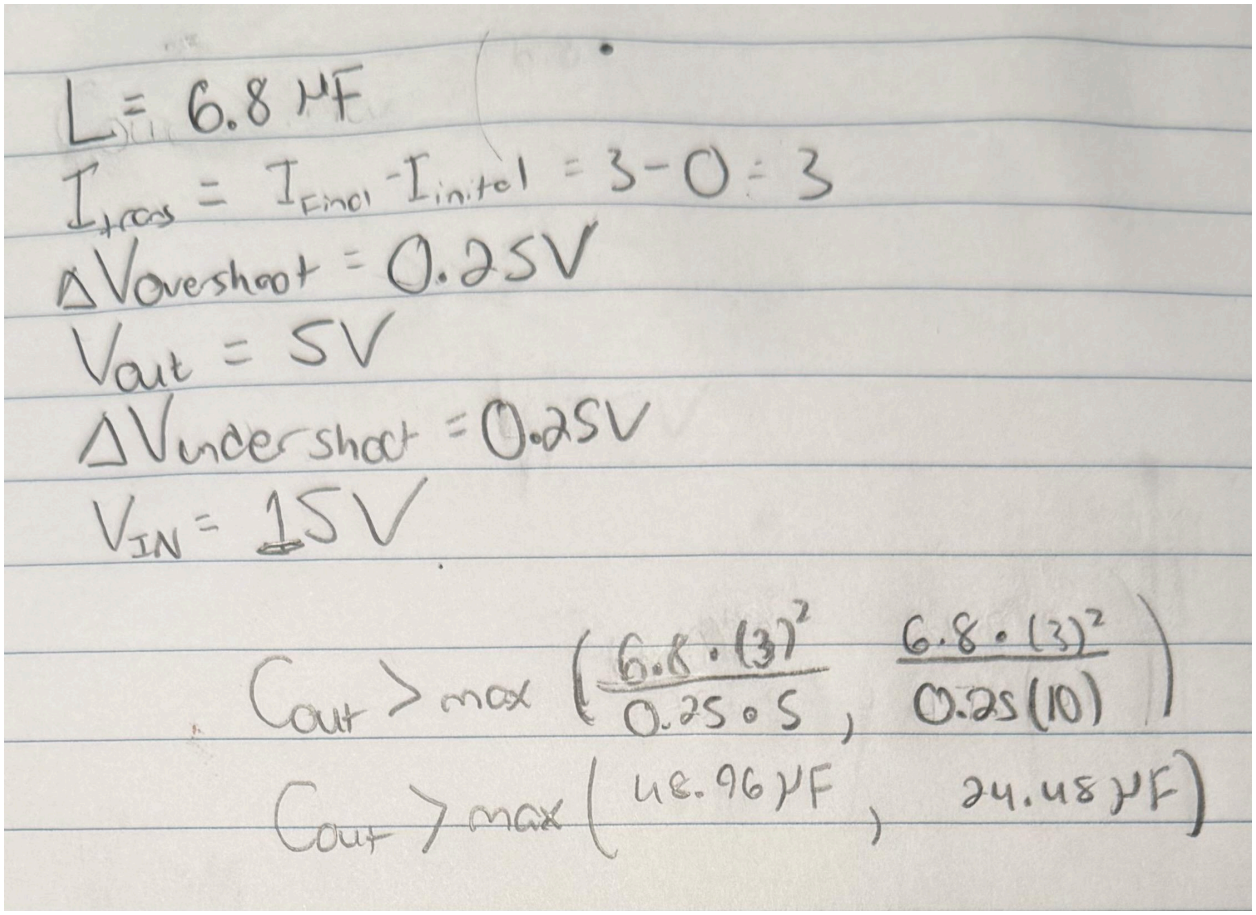
Output Capacitor

The output capacitor's main function is to keep the output voltage ripples small, ensure feedback loop stability, and keep voltage in check during sudden changes in current. When current load decreases the inductor continues to supply current which charges the capacitor. When current load increases the output capacitor provides excess current before the inductor. For the output capacitor I chose the CC0805MFX5R5BB226 a 22 µF 6.3V ceramic capacitor. The main things I looked for when choosing this capacitor was a low ESR and a capacitor that could withstand over 5V.

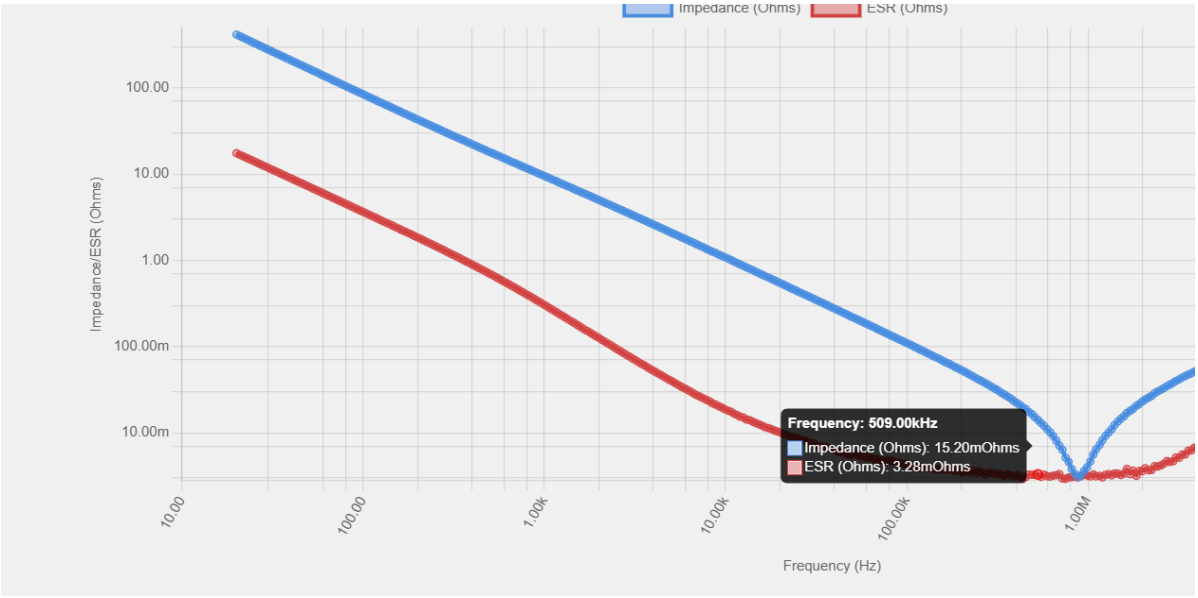


The DC Bias on the capacitor was about 65.53% which means a 22 µF capacitor provides 7.70 µF at 5V.

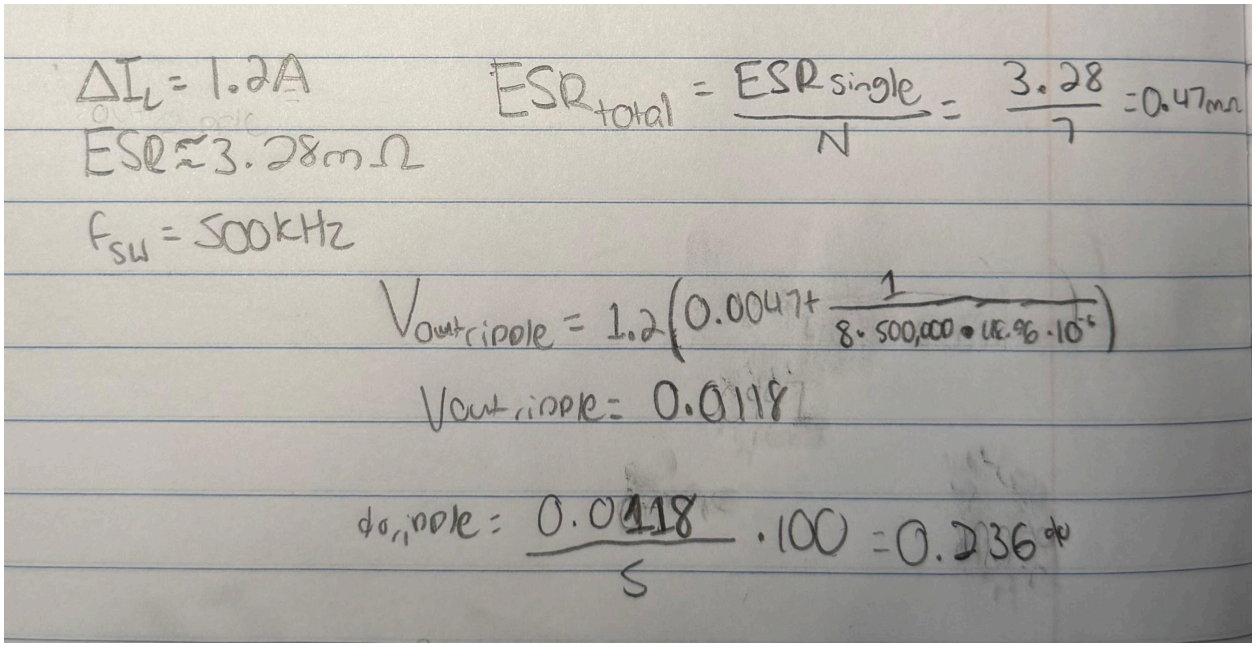
$$\mathbf{COUT} > \max\left(\frac{\mathbf{L} \cdot \mathbf{I}_{\mathbf{Trans}}^2}{\Delta\mathbf{V}_{\mathbf{Overshoot}} \cdot \mathbf{VOUT}}, \frac{\mathbf{L} \cdot \mathbf{I}_{\mathbf{Trans}}^2}{\Delta\mathbf{V}_{\mathbf{Undershoot}} \cdot (\mathbf{VIN} - \mathbf{VOUT})}\right)$$



Using the Cout formula I was able to see how much capacitance I needed from the capacitors. In total I needed 48.96 µF. With one 22 µF capacitor at 5V equivalent to 7.70 µF I calculated I would need seven capacitors in parallel to reach that 48.96 µF mark. With seven capacitors it would be a total of 53.9 µF. With a 20% margin of error this is a little under the 48.96 µF mark , but I still opted to choose this capacitor.



$$\mathbf{VOUT_{Ripple}} = \mathbf{\Delta I_L} \cdot \left(\mathbf{ESR} + \frac{\mathbf{1}}{\mathbf{8 \cdot f_{sw} \cdot COUT}}\right)$$



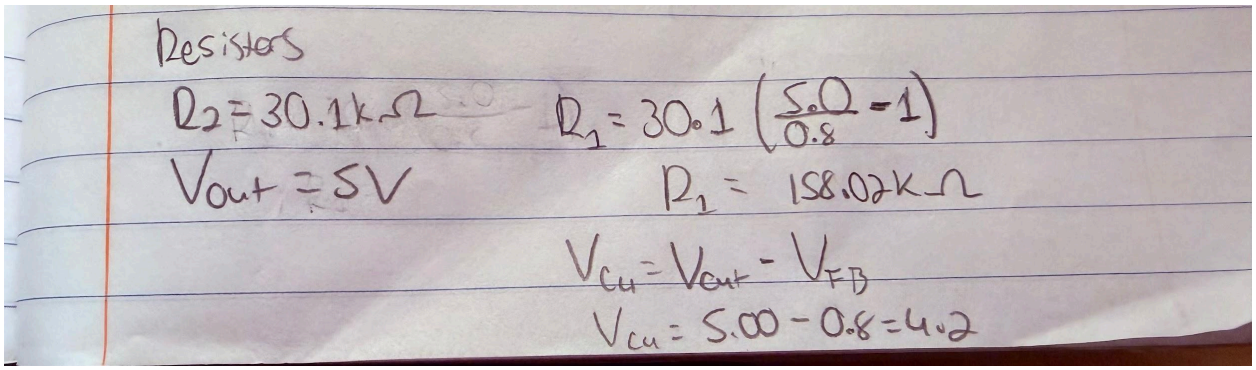
The ESR on the capacitor is important as it decides how much voltage ripple the component has. At 500Khz the ESR is about 3.28mΩ. With seven capacitors in parallel this ESR becomes 0.47mΩ. Using this I calculated the voltage ripple of 0.236%.

Bootstrap Capacitor

The bootstrap capacitor is there to make sure Q1 is always provided with the necessary voltage to stay on. I used the recommended 0.01 µF value given by the manufacturer and made sure the capacitor could withstand more than 2.85V. I went with the C0603C104K8RACTU which is rated at 0.01 µF and 10V.

Resistors

$$\mathbf{R1 = R2 \cdot \left(\frac{VOUT}{0.8V} - 1\right)}$$



The resistors create a voltage divider that makes sure the component gets 5V and allows the IC to compare the FB pin to its 0.8V reference. The IC is able to adjust voltage if the FB pin isn't the same voltage as the reference. I went with the recommended values again for the resistors as they were used to get a Vout of 5V. I used the equation to double check the values. I chose the RC0603FR-07158KL, a 158kΩ resistor and the RC0603FR-0730K1L, a 30.1kΩ resistor.

Transient Capacitor

To improve transient response I chose the 885012006021 a 47pF capacitor. This was a recommended value by the manufacturer.

Cost of Components

Component	Cost	Link	Units	Size
Buck IC	$0.96	https://www.digikey.com/en/produc	1	0.114" L × 0.110" W (2.90 mm × 2.80 mm)

		ts/detail/diodes-incorporated/AP63301WU-7/10491607		
Inductor	$0.65	https://www.digikey.com/en/products/detail/bourns-inc/SRN8040TA-6R8M/6155128	1	0.315" L x 0.315" W (8.00mm x 8.00mm)
Input Capacitor	$3.40	https://www.digikey.com/en/products/detail/tdk/C2012X7S1E106K125AE/10413498	4	0.079" L x 0.049" W (2.00mm x 1.25mm)
Output Capacitor	$3.15	https://www.digikey.com/en/products/detail/yageo/CC0805MFX5R5BB226/21767714?curr=usd&utm_campaign=buynow&utm_medium=aggregator&utm_source=octopart	7	0.079" L x 0.049" W (2.00mm x 1.25mm)
Bootstrap capacitor	$0.16	https://www.digikey.com/en/products/detail/kemet/C0603C104K8RACTU/411094	1	0.063" L x 0.032" W (1.60mm x 0.81mm)
Resistor # 1	$0.10	https://www.digikey.com/en/products/detail/yageo/RC0603FR-07158KL/726963	1	0.063" L x 0.031" W (1.60mm x 0.80mm)
Resistor # 2	$0.10	https://www.digikey.com/en/products/detail/yageo/RC0603FR-0730K1L/727142	1	0.063" L x 0.031" W (1.60mm x 0.80mm)
Transient	$0.10	https://www.digik	1	0.063" L x 0.031" W

Capacitor		ey.com/en/products/detail/w%C3%BCrth-elektronik/885012006021/5453619		(1.60mm x 0.80mm)
Total Cost	$8.52			

Works Cited

TDK Corporation. C2012X7S1E106K125AE: Multilayer Ceramic Chip Capacitors. TDK, https://product.tdk.com/system/files/dam/doc/product/capacitor/ceramic/mlcc/charasheet/c2012x7s1e106k125ae_211020.pdf

Diodes Incorporated. AP63300/AP63301: 3.5A, 32V Synchronous Buck Converter. Diodes Incorporated, https://www.diodes.com/assets/Datasheets/AP63300-AP63301.pdf

YAGEO Corporation. “CC0805MFX5R5BB226.” YAGEO, https://yageogroup.com/products/Capacitors/part/CC0805MFX5R5BB226

Monolithic Power Systems. “Step-by-Step Design Process.” Buck Converter Topology Design Guide, Monolithic Power Systems, https://www.monolithicpower.com/en/buck-converter-topology-design-guide/design-steps-and-considerations/step-by-step-design-process